

MM — core momentum, rebuild candidate

Nifty 250 · volatility-adjusted 12-month momentum · 25 names · monthly · one order day per month · research report generated 2026-09-10 from the honest master store · backtest, not live; net of 20 bps slippage each way; price-return basis (dividends excluded from portfolio and benchmarks) · run 3e7c32fe8a · not a client document

Read first — what is in-sample here. The universe, score, lookback, skip, position count, exit buffer and monthly cadence were chosen on 2010-2015 before out-of-sample was opened on 2026-09-10. **Everything else — the trailing stop, inverse-volatility sizing, the hold-15-in-bear rule, the regime detector and the sector cap — was chosen after the 2016-2026 window had been seen**, across roughly 128 post-OOS cells. Each element rests on a plateau rather than a single cell, but **this stack as a whole has never been judged on unseen data**. The 2016-2026 figures below are therefore not a clean out-of-sample result for the stack; they are the window the later devices were selected on.

Every number is a simulation on reconstructed point-in-time index membership with delisted names carried to their last price. Tax figures use a flat 25% on each financial year's realised gain with no loss carry-forward, which is deliberately harsh.

2016-26 (out-of-sample)

CAGR 25.7% post-tax 19.9%	Total return 1049% Nifty 250 306%	Max drawdown -26.5% 324 days peak to recovery	Sharpe (rf 5%) 1.18 Nifty 250 0.56	Sortino 1.49 Calmar 0.97	Volatility 17.5% Nifty 250 16.2%	Hit rate 52.7% 624 closed trades	Avg holding 129 days median 91	Order days 1.0 / month max 1 in any month			
	CAGR	Total	Vol	Sharpe	Sortino	Calmar	Max DD	DD days	Best month	Worst month	Positive months
Portfolio	25.7%	1049%	17.5%	1.18	1.49	0.97	-26.5%	324	16.2%	-11.7%	66%
Portfolio post-tax	19.9%	596%	18.3%	0.81	0.95	0.75	-26.5%	350	16.2%	-11.7%	64%
Nifty 250	14.0%	306%	16.2%	0.56	0.64	0.36	-38.5%	1020	14.6%	-25.6%	66%
Nifty 500	12.1%	240%	16.1%	0.44	0.52	0.32	-38.3%	294	14.5%	-24.2%	62%

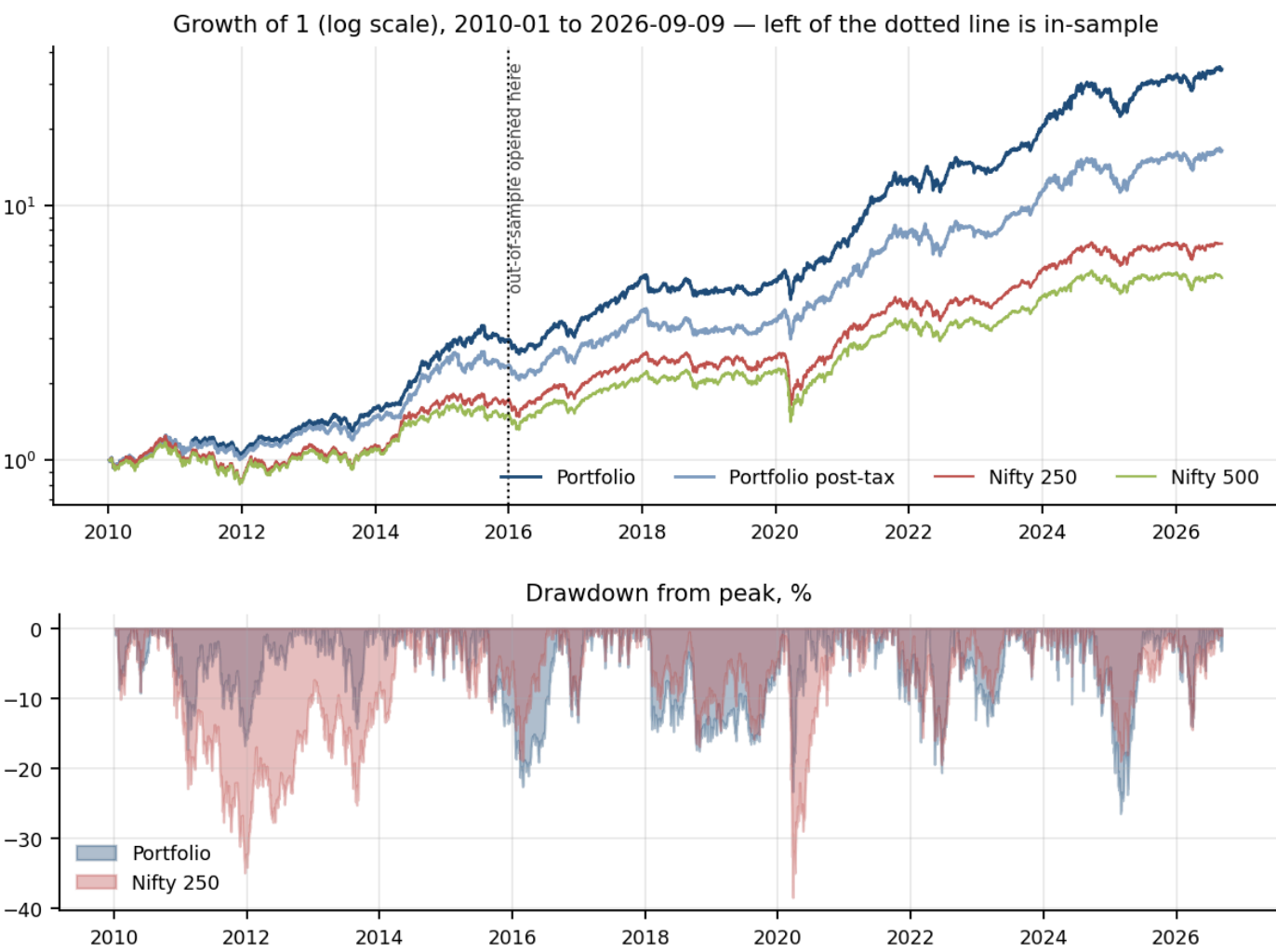
2010-26 (full span, includes the in-sample window)

CAGR 23.6% post-tax 18.3%	Total return 3328% Nifty 250 608%	Max drawdown -26.5% 324 days peak to recovery	Sharpe (rf 5%) 1.15 Nifty 250 0.46	Sortino 1.47 Calmar 0.89	Volatility 16.2% Nifty 250 16.2%	Hit rate 52.6% 896 closed trades	Avg holding 139 days median 92	Order days 1.0 / month max 1 in any month			
	CAGR	Total	Vol	Sharpe	Sortino	Calmar	Max DD	DD days	Best month	Worst month	Positive months
Portfolio	23.6%	3328%	16.2%	1.15	1.47	0.89	-26.5%	324	16.2%	-11.7%	68%
Portfolio post-tax	18.3%	1543%	17.0%	0.78	0.92	0.69	-26.5%	350	16.2%	-12.9%	64%
Nifty 250	12.5%	608%	16.2%	0.46	0.57	0.32	-38.5%	1020	14.6%	-25.6%	62%
Nifty 500	10.4%	420%	16.1%	0.33	0.42	0.27	-38.3%	294	14.5%	-24.2%	60%

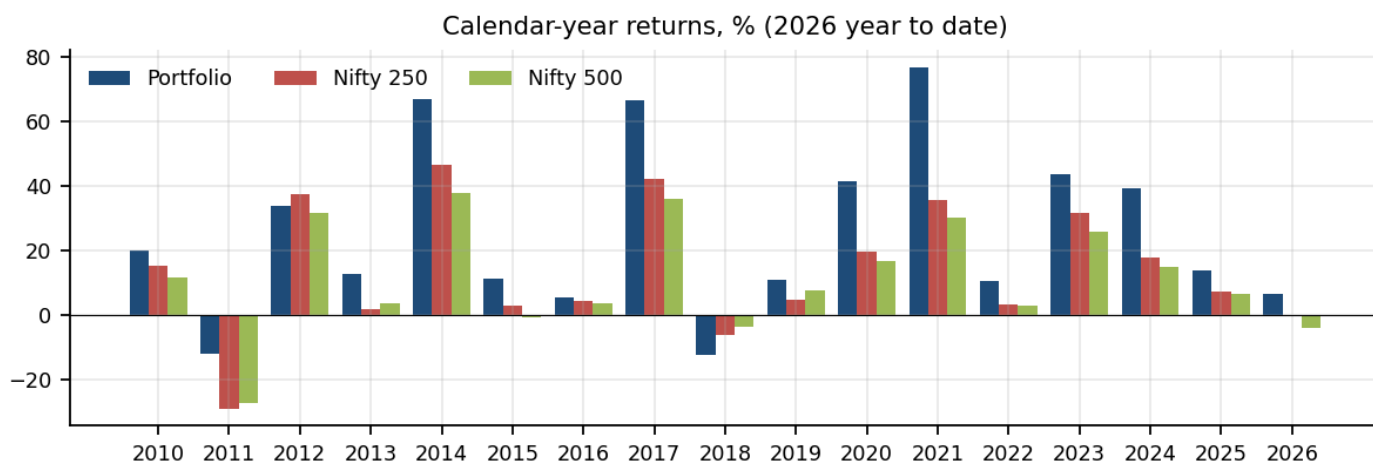
Period returns to 2026-09-09

Period returns (annualised beyond 1Y)	1M	3M	6M	YTD	1Y	3Y	5Y	10Y	Since 2010
Portfolio	-0.6%	6.8%	13.4%	6.5%	12.2%	24.9%	23.6%	26.0%	23.6%
Portfolio post-tax	-0.6%	6.8%	10.0%	3.3%	8.9%	18.6%	17.4%	19.8%	18.3%
Nifty 250	-0.8%	5.0%	7.8%	0.1%	4.7%	11.9%	11.9%	13.7%	12.5%
Nifty 500	-3.3%	2.6%	4.1%	-3.8%	0.0%	9.5%	9.1%	11.7%	10.4%

Growth and drawdown



Calendar years



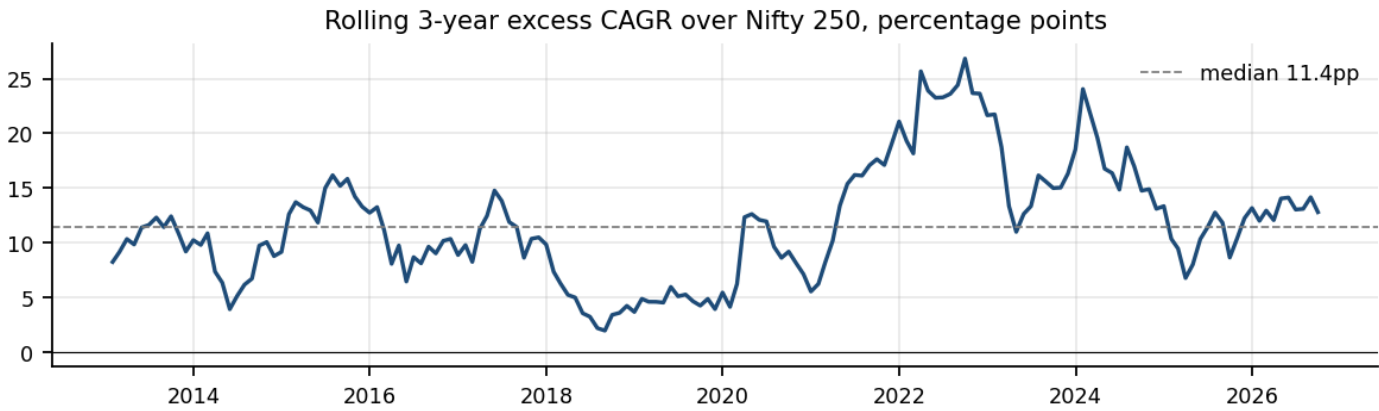
Year	Portfolio	Portfolio post-tax	Nifty 250	Nifty 500	Excess vs Nifty 250
2010	20.1%	19.9%	15.5%	11.9%	4.6%
2011	-11.9%	-15.0%	-28.9%	-27.2%	17.1%
2012	34.0%	32.9%	37.4%	31.8%	-3.4%
2013	12.8%	9.0%	1.8%	3.6%	11.0%
2014	67.1%	60.9%	46.6%	37.8%	20.6%
2015	11.5%	-0.7%	2.9%	-0.7%	8.6%
2016	5.5%	5.5%	4.6%	3.8%	0.9%
2017	66.5%	55.1%	42.4%	35.9%	24.1%
2018	-12.4%	-16.6%	-6.2%	-3.4%	-6.2%
2019	10.9%	10.5%	5.0%	7.7%	5.9%
2020	41.5%	41.5%	19.7%	16.7%	21.8%
2021	76.9%	57.1%	35.7%	30.2%	41.3%
2022	10.7%	0.0%	3.4%	3.0%	7.4%
2023	43.6%	43.1%	31.6%	25.8%	12.1%
2024	39.2%	25.3%	17.7%	15.2%	21.5%
2025	14.0%	12.0%	7.3%	6.7%	6.8%
2026	6.5%	3.3%	0.1%	-3.8%	6.5%

Monthly returns, % (portfolio, pre-tax)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2010	-5.9	2.7	4.2	3.0	-4.1	1.4	3.2	3.0	6.4	4.2	0.6	0.5	20.1
2011	-5.8	-4.9	9.4	2.5	-0.6	0.4	1.0	-5.7	-1.3	1.8	-5.2	-3.1	-11.9
2012	6.0	3.3	4.9	0.6	-0.9	0.3	-2.1	3.5	5.0	2.3	4.0	3.1	34.0
2013	-0.9	-1.0	0.5	2.1	4.7	-6.1	-0.8	-5.0	6.9	6.3	1.0	5.2	12.8
2014	-1.1	5.1	-1.0	0.8	6.8	11.4	4.3	7.1	10.0	1.4	4.8	3.8	67.1
2015	5.9	1.3	2.1	-2.1	5.9	1.2	5.3	-3.7	-2.1	1.3	-3.5	0.1	11.5
2016	-5.2	-7.5	3.7	5.1	-1.8	6.6	6.6	5.5	-0.2	2.9	-5.3	-3.6	5.5
2017	11.8	3.8	2.4	5.2	1.4	2.0	4.8	3.1	0.6	8.6	2.1	7.0	66.5
2018	-4.6	-6.5	-1.2	7.1	-4.4	-0.8	3.3	5.9	-8.4	-2.9	1.3	-0.7	-12.4
2019	1.2	-0.4	1.1	1.2	0.9	-0.0	-5.1	0.0	2.8	6.5	0.1	2.5	10.9
2020	4.0	-1.7	-11.3	16.2	-2.2	9.5	-1.0	5.9	3.0	0.9	10.4	4.3	41.5
2021	-4.2	11.5	8.6	9.9	9.2	5.7	0.6	9.0	5.6	-1.3	0.3	4.8	76.9
2022	-0.8	-5.6	14.1	-1.7	-5.8	-3.2	10.5	6.7	1.1	0.2	1.0	-4.1	10.7
2023	-0.1	-4.7	2.8	1.7	7.5	7.0	3.8	0.5	3.6	-3.5	8.8	10.4	43.6
2024	10.9	-0.4	2.8	6.6	3.5	6.0	11.8	0.9	-1.3	-7.0	-1.5	2.6	39.2
2025	-10.2	-11.7	11.1	4.9	8.0	7.6	1.4	-2.1	-1.2	4.2	4.1	-0.0	14.0
2026	-3.0	0.1	-9.7	12.3	4.4	0.9	1.9	2.7	-1.9				6.5

Largest drawdowns

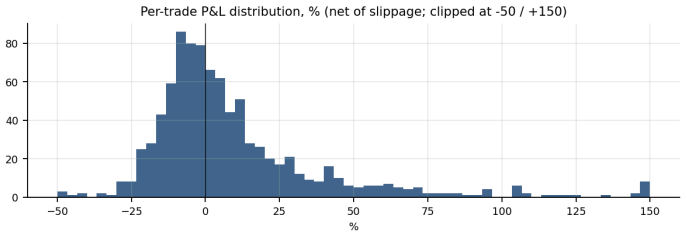
Peak	Trough	Recovered	Depth	Days down	Days to recover	Total days
2024-09-02	2025-02-28	2025-07-23	-26.5%	179	145	324
2020-02-24	2020-03-23	2020-06-04	-23.4%	28	73	101
2015-08-20	2016-02-26	2016-09-06	-22.7%	190	193	383
2022-04-12	2022-06-20	2022-08-12	-20.6%	69	53	122
2018-01-24	2018-10-25	2020-02-05	-17.6%	274	468	742
2010-11-11	2011-02-09	2012-09-10	-17.3%	90	579	669



Trades

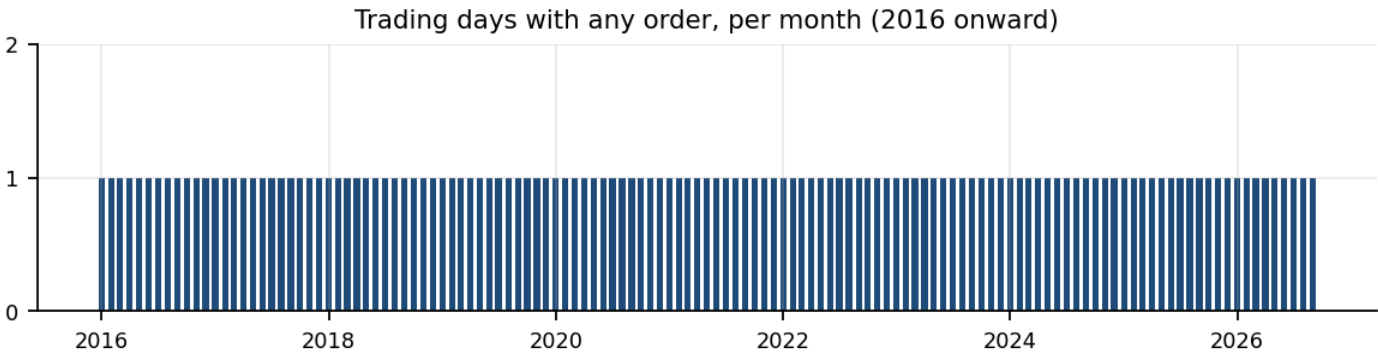
Trade statistics	2010-26	2016-26 (OOS)
Closed trades	896	624
Win rate	52.6%	52.7%
Average winner	29.5%	28.4%
Average loser	-10.5%	-11.0%
Expectancy per trade	10.5%	9.8%
Profit factor (sum of wins / sum of losses)	3.11	2.87
Best trade	871.5%	541.2%
Worst trade	-65.8%	-57.9%
Average holding days	139	129
Median holding days	92	91
Exits by trailing stop	23.1%	24.7%
Trades per year	110	118
One-way turnover per year	2.3x	2.3x
Slippage cost per year	0.9%	0.9%
Order days per month	1.0	1.0
Months with any order	99%	100%

Holding period	Share of trades	Avg P&L
0-30 days	10.7%	-3.3%
31-60 days	16.5%	-1.6%
61-120 days	29.1%	-2.0%
121-250 days	29.0%	10.0%
251-5000 days	14.6%	60.4%



Followability — one order day a month

The stack rebalances and checks the stop on the same monthly signal, so every order falls on one day. Across 2016-26 that is **1.0 trading days with any order per month**, never more than 1, with orders in 100% of months. Faster rhythms were tested and cost performance: biweekly rebalancing takes the Sharpe from 1.18 to about 1.0 and weekly to about 1.02, both at two to four order days a month (\$10). Engagement between rebalances is a product question — a weekly note reporting rankings, distance to stops and regime state without trading — not a strategy one.



Behaviour against the benchmark

Up- and down-capture are computed on monthly returns against the Nifty 250 throughout this table. A down-capture below 1 means the book falls less than the index in down months.

Window	Portfolio CAGR	Nifty 250 CAGR	Portfolio Sharpe	Nifty 250 Sharpe	Portfolio Max DD	Up-capture	Down-capture
2016-26 (OOS)	25.7%	14.0%	1.18	0.56	-26.5%	1.20	0.77
2016-19	14.3%	9.9%	0.71	0.36	-17.6%	0.95	0.70
2020-22	40.4%	18.8%	1.73	0.65	-23.4%	1.24	0.64
2023-26	27.3%	14.6%	1.18	0.69	-26.5%	1.41	1.01
Wright window Oct-20 to Aug-26	33.9%	19.2%	1.50	0.96	-26.5%	1.38	1.00

Against Wright Momentum's published grid

Capture here is measured against the same synthetic mid-small 400 series used in §9-§10, so it is comparable to Wright's published figures. Theirs are gross of costs, this book is net of 20 bps each way, and Wright publish no Sharpe for this window. Their drawdown is measured on month-end values and is the shallower of the two on any measure; on a daily basis this book's is deeper still relative to theirs. The gap in return is 2026 and the last three years; the gap in drawdown is theirs to keep.

Oct-2020 to Aug-2026, vs mid-small 400	CAGR	Sharpe	Max DD	Up-capture	Down-capture
This book (net of 20 bps)	33.9%	1.50	-26.5%	1.09	0.73
Wright Momentum (published, gross)	31.2%	not published	-21.7% (month-end)	1.11	0.90

The stack

Parameter	Value	Decided in
Universe	Nifty LargeMidcap 250, point-in-time	§1 (in-sample)
Score	volatility-adjusted momentum, 5% vol floor	§1 (in-sample)
Lookback	252 sessions (12m), min 219	§1 (in-sample)
Skip	21 sessions	§3a (in-sample)
Positions	25	§2 (in-sample)
Exit buffer	20 (sell below rank 45)	§2 (in-sample)
Rebalance	monthly, first trading day	§2 (in-sample)
Trailing stop	20% from peak	§3c rejected in-sample, adopted §8 post-OOS
Stop check	monthly, same signal as the rebalance	§10 (post-OOS)
Sizing	inverse volatility, 63d window, 10% cap	§9b (post-OOS)
Bear book	hold at most 15 names, bear exit buffer 20	§9b/§9c (post-OOS)
Regime	NIFTY 100 ROC 31, 3-day confirm, lagged	§9b (post-OOS)
Sector cap	5 names per NSE sector at entry	§9d (post-OOS)

The bear rule is the distinguishing device: when the regime is bear the book holds at most 15 names instead of 25 and lets the exit rank widen to 35, which is where the down-capture comes from. Because the engine never resizes an existing position, names bought at 1/25 in a bull regime are not topped up in bear, so the bear book carries some cash — it averages 17 names at about 77% invested rather than a full 15-name book.

Gates (pre-committed 2026-09-10, signed before the first search)

Gate	Value	Status
G1 IS Sharpe 2010-2015 ≥ 0.8	1.10	pass
G2 OOS Sharpe 2016-26 ≥ 0.9	1.18	pass
G3 each sub-window ≥ 0.6	0.71 / 1.73 / 1.18	pass
G4 OOS max drawdown no worse than −40%	-26.5%	pass
G5 walk-forward within 0.2 of static OOS	5y refit 0.98 (gap 0.200) · 3y refit 0.97 (gap 0.215)	ON THE LINE
G6 parameter count ≤ 10	13 listed	OVER — see note
G7 turnover reported	2.3x one-way per year, 118 trades	reported
G8 OOS CAGR ≥ 20% pre-tax	25.7%	pass

Two gates are not clean. G6 caps the all-in parameter count at 10 and the table above lists 13; whether the vol floor, minimum observations and the regime's two settings count is a judgement, but on any reading this stack is at or over the limit. G5 was run in §11: refitting the post-OOS devices yearly on a trailing five years gives Sharpe 0.98, a gap of 0.200 — the threshold to three decimals — and on a trailing three years 0.97, a gap of 0.215, which misses. The stack sits on the line rather than clearing it.

What to expect, rather than the headline

The adopted cell is **rank 1 of the 32 cells** in §11's refit grid, which runs 0.88 to 1.18 with a median of 1.04. That is what selecting a cell on the window it is measured on produces. Four readings of the same book, from most to least flattering:

Reading	CAGR	Sharpe	Source
Static, the adopted cell (selection-window maximum)	25.7%	1.18	rank 1 of 32
Centre of the refit-grid plateau	—	1.04	median of 32 cells
Walk-forward, 5-year refit (no hindsight)	22.4%	0.98	§11
Walk-forward, 3-year refit	22.0%	0.97	§11
Deflated for the 32-cell refit grid	—	0.76	Bailey-Lopez de Prado
Deflated for the ~128 post-OOS cells of §8-§10	—	0.66	Bailey-Lopez de Prado

The walk-forward row is the one to plan against — roughly 22% a year at a Sharpe near 1.0, not 25.7% at 1.18. The deflated rows apply the Bailey-López de Prado haircut for the number of post-OOS cells examined, using the observed cross-trial Sharpe variance of 0.041 rather than an assumed one.

Tax and SIP

Portfolio taxed 25% on each financial year's positive gain at 31 March, since the turnover realises gains yearly. Indices taxed 25% once at exit, as a buy-and-hold investor would be. Post-tax CAGR 2016-26: portfolio 19.9% against Nifty 250 14.0% pre-tax.

SIP Rs 10,000 / month	Invested	Portfolio (value, XIRR)	Nifty 250	Nifty 500
From 2010, pre-tax	20.1L	235.0L (11.7x), 25.7%	73.6L (3.7x), 14.1%	59.3L (3.0x), 11.9%
From 2010, post-tax	20.1L	131.1L (6.5x), 19.9%	73.6L (3.7x), 14.1%	59.3L (3.0x), 11.9%
From 2016, pre-tax	12.9L	62.6L (4.9x), 27.7%	29.2L (2.3x), 14.6%	25.4L (2.0x), 12.2%
From 2016, post-tax	12.9L	43.5L (3.4x), 21.4%	29.2L (2.3x), 14.6%	25.4L (2.0x), 12.2%

Caveats

- The stack is a post-OOS design. Its 2016-26 numbers are selection-window numbers, not out-of-sample ones; the honest out-of-sample test is the next few years of live running.
- Roughly 128 post-OOS cells were examined across §8-§10; the deflation that implies is not applied to the figures above.
- Benchmarks are price indices. Nifty 250 is the real index from 2020 spliced onto a 50/50 NIFTY 100 + Midcap 150 quarterly-rebalanced proxy.
- Sector labels come from 26 archived NSE constituent lists and cover 87% of all-ever members; 13% are unlabelled and unconstrained by the cap, about 1% of the book's buys.
- No exposure overlay. 2008-type events sit outside the tested span.
- Capacity, market impact beyond 20 bps, and the tax treatment of an actual account are not modelled.